

FreshGuard

End-User User Guide

Zero Food Waste | Smart IoT Revolution



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Practical guide to monitoring food storage, freshness indicators, alerts, and inventory

For household and small-restaurant users

Project	FreshGuard
Theme	Zero Food Waste
Category	Smart IoT Revolution
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Purpose	User-focused guidance for monitoring storage conditions, food records, freshness indicators, alerts, and waste-reduction routines.
Version	FreshGuard SRS v1.0-aligned user guide

Scope — Use this guide for normal operation in a household refrigerator or small-restaurant storage area, including an insulated box or cooler simulation where applicable.

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2. About FreshGuard

FreshGuard is an IoT food-storage monitoring system designed to help households and small restaurants see what food is stored, observe storage conditions, recognize freshness-related statuses, and respond to alerts. It brings local device indicators and dashboard information together so users can make more informed storage and inventory decisions.

FreshGuard supports visibility, monitoring, status indication, alerts, and record management. It does not independently certify food as safe, replace product labels, or remove the need for human inspection and applicable food-safety practices.

Key idea — Use FreshGuard as a practical monitoring aid that helps reduce avoidable waste while keeping human judgment and food-safety guidance at the center of every decision.

3. Purpose and What the System Does

FreshGuard records useful information about stored food, observes configured storage conditions, tracks time and expiry-related information, shows freshness statuses, and presents local and dashboard information. It can raise alerts when configured conditions or record information require attention.

Typical user outcomes

- Record food names, dates, categories, locations, and unique identifiers.
- Review temperature, humidity, door state, and relative air-quality or gas-change indications where available.
- See Fresh/Normal, Use Soon, or Check Food statuses based on the implemented configuration.
- Receive local and, where configured, dashboard or remote notifications.
- Update records when food is moved, opened, consumed, discarded, or otherwise changed.

Important — FreshGuard supports decisions; it does not replace human inspection, product labels, approved food-safety practices, or disposal procedures.

4. Who Should Use It and Scope

This guide is for household users and small-restaurant staff who need a simple way to monitor stored food and keep inventory information current. It assumes that users can safely place containers, observe the device, review the dashboard when connected, and follow local food-safety procedures.

This guide covers

- Normal operation and initial setup.
- Food registration, record updates, and optional QR identification.
- Local indicators, dashboard information, freshness statuses, and alerts.
- Offline operation, reconnection, troubleshooting, and safe use.

Outside the scope – source code, pin maps, credentials, backend implementation, controller programming, and other developer topics are not covered. The exact hardware arrangement and available fields depend on the implemented FreshGuard setup.

5. System Overview and Main Parts

FreshGuard combines sensing, local feedback, records, and dashboard views. The exact hardware arrangement depends on the implemented setup, so use the approved setup instructions for the device supplied to you.

User-facing parts may include

Part	User purpose
Controller or device	Collects readings and coordinates local indicators and dashboard updates.
Temperature and humidity sensors	Show storage-environment readings for the configured location.
MQ-135 or equivalent sensor	Indicates relative changes in air-quality or gas patterns.
Door/open-state sensor	Helps identify an open or improperly closed storage door.
LCD, LEDs, and buzzer	Provide local readings, statuses, prompts, and alerts.
Dashboard	Shows readings, records, statuses, alerts, reports, and visualized data.
Food records and inventory	Store identity, dates, category, location, and status information.
Optional QR identification	Helps find an assigned food record when QR functionality is implemented.

Simple flow: Sensors → FreshGuard device/controller → Local indicators and dashboard → User action

6. Getting Started and Initial Setup

1. Choose a suitable refrigerator, storage area, or insulated box/cooler simulation for the implemented setup.
2. Place the device and sensors safely, following the approved placement instructions.
3. Connect the approved power source. Keep power connections protected from liquids.
4. Power on the unit and allow readings to stabilize according to the implemented setup.
5. Confirm that the LCD, LEDs, and any other local indicators operate as expected.
6. Connect to the configured network if dashboard access is required. Do not guess credentials or setup values.
7. Open the dashboard and confirm that the device is visible when the network is available.
8. Create initial food records using the information available on labels or approved local records.
9. Verify that readings, food records, and statuses appear in the expected local or dashboard view.

Tip — Complete setup before storing important food, then perform one test review from the device and one from the dashboard when connected.

7. Safe Placement of Sensors and Food Containers

Placement affects readings and the usefulness of alerts. Follow the implemented setup instructions and use these practical checks:

- Keep sensors away from direct contact with food or liquid.
- Avoid blocked airflow, heavy condensation, spills, sharp edges, and unnecessary heat sources.
- Secure cables so they cannot be pulled, pinched, or caught in a door.
- Leave enough room for air circulation around containers and sensors.
- Separate raw and ready-to-eat foods using suitable locations and closed containers.
- Use closed or otherwise appropriate containers for the stored food.
- Arrange food so the door can close fully without pressing on containers or cables.

Safety — Do not place electronics where they can be splashed, submerged, crushed, or exposed to unsafe heat. If placement instructions conflict with a local condition, stop and use the approved support route.

8. Powering On and Understanding the Device

Before powering on, inspect the visible connections and confirm that the device, sensors, and cables are positioned safely. Use only the approved power source and do not handle wiring or power connections if doing so would create a hazard.

After startup, observe the LCD, LEDs, and buzzer if present. Wait for readings to become available according to the implemented setup, then check whether the local information agrees with the dashboard when connected.

Do not guess from an indicator alone. A warning, blank display, missing reading, or unavailable status requires the relevant troubleshooting step. It must not be treated as proof that food is safe or unsafe without human review.

9. Food Registration

Create a record for each food item or storage unit that the implemented setup is intended to track.

1. Enter the food name.
2. Enter the manufacture date when it is available.
3. Enter the expiry or best-before date when it is available.
4. Assign or enter the unique ID used by the local setup.
5. Select the food category.
6. Add storage information such as location, container, or category-specific notes available in the setup.
7. Review spelling, dates, category, and identifier before saving.
8. Save the record and confirm that it appears in the inventory.
9. Update the record when the food is moved, opened, consumed, or discarded.

Important — Do not invent a manufacture, expiry, or best-before date. If packaging information is unavailable, use the approved local procedure or a notes field if one exists.

10. QR Code Identification and Updating Food Records

QR identification is optional and is available only where it has been implemented in the deployment. When available:

1. Open the QR function in the approved user interface.
2. Scan the food label or assigned code.
3. Confirm that the matched name and unique ID are correct.
4. Review the dates, category, location, and current status.
5. Update the record after the food is moved, opened, consumed, or discarded, then save the change.

Reminder — A successful scan identifies a record; it is not proof that the food is safe. If QR is unavailable or not implemented, search by unique ID or name and update the record normally.

11. Understanding the Freshness Statuses

Status	Meaning and user action
Fresh/Normal	Current recorded conditions and timing are within the configured expected range for that food category. Continue normal checks and follow product labels.
Use Soon	The record or monitored conditions indicate that attention is needed soon. Prioritize use, verify dates and storage, and inspect before use.
Check Food	A condition, data pattern, timing issue, or unavailable or abnormal reading requires human checking. Inspect the food and storage conditions and follow food-safety guidance.

Limitation — Status logic and thresholds are configured for the implemented setup and may vary by food category. A status is an attention aid, not a guaranteed spoilage decision.

12. Understanding the Dashboard

Depending on the implemented setup, the dashboard may show live sensor readings, food inventory, manufacture and expiry or best-before information, storage duration, freshness status, alerts, reports, data visualizations, and individual record details.

Read the dashboard in this order

1. Check the connection state and the time of the latest data.
2. Review new or urgent alerts.
3. Review sensor readings and any unavailable fields.
4. Review inventory statuses, dates, and storage duration.
5. Open the affected food record or storage location.
6. Take the appropriate action and update the record.

Displayed fields, report views, and data refresh behavior depend on the implemented setup. Use the device indicators when dashboard data is delayed or unavailable.

13. Understanding LCD and LED Indicators

The LCD may provide local readings, freshness statuses, prompts, or alerts according to the implemented setup. Use the message shown on the LCD to understand what needs attention, then check the dashboard or the relevant record when available.

Indicator	User-level meaning
Green	Normal operation or no immediate attention indicated.
Yellow	Caution or attention soon, commonly associated with Use Soon or a warning.
Red	Urgent attention, an abnormal condition, Check Food, or an alert requiring prompt inspection.

Context matters — Exact LED logic can be configured. Check the LCD message and dashboard status rather than relying on color alone. Do not infer meaning from blink rates.

14. Temperature and Humidity Monitoring

Temperature and humidity readings describe conditions in the monitored storage environment. View them on the LCD when available and on the dashboard when the device is connected. Stable sensor placement and unobstructed airflow help make readings more useful.

When a reading looks unusual, work through these checks:

- Confirm that the door closes fully and has not been left open.
- Check sensor placement, airflow, condensation, and nearby containers.
- Check power and whether the displayed reading is current.
- Review the affected food category and storage record.
- Inspect the food and follow product labels and food-safety guidance.

Important — Expected ranges depend on food category and implemented configuration. This guide does not provide universal temperature or humidity thresholds.

15. Gas/Air-Quality Monitoring

The MQ-135 or equivalent sensor observes relative changes in air-quality or gas patterns in the storage environment. A change can prompt you to inspect the storage area and food, but it is only an indication of environmental change.

The reading does not by itself confirm spoilage, identify a specific contaminant, or replace smell, appearance, label checks, human judgment, or food-safety guidance. Do not consume or discard food solely because of one gas or air-quality reading.

At user level, remember that the sensor may need an initial warm-up, suitable placement, careful handling, and the configuration used by the implemented setup. Readings can be affected by airflow, condensation, nearby materials, and configuration limitations. Follow the approved setup instructions and investigate unusual changes with the other available evidence.

16. Door Monitoring and Door-Open Alerts

The door/open-state sensor helps identify an open or improperly closed refrigerator, box, or cooler door. It supports awareness of storage interruptions but does not measure food safety directly.

1. Close the door fully and check that the seal or closure is not obstructed.
2. Check that containers, cables, or sensor parts are not preventing closure.
3. Wait for the system to update the door state.
4. If the alert continues, review the LCD or dashboard message and inspect the sensor placement.

Repeated door alerts may be caused by alignment, placement, an obstruction, or a sensor or setup issue. Keep the door closed while investigating and escalate if the alert remains after safe user-level checks.

17. Alerts and Notifications

FreshGuard may provide local alerts through the buzzer, LCD, and LEDs. The dashboard may also show alerts and affected records. An approved remote notification service may be available when that service is implemented and configured for the deployment.

Use local monitoring even when remote notifications are enabled. Remote notifications depend on network and service availability, and delivery cannot be assumed when the

network is unavailable or delayed. Do not infer service names, credentials, message formats, or guaranteed delivery from an alert screen.

Good practice — Treat every alert as a prompt to investigate the storage condition and affected record, not as an automatic instruction to consume or discard food.

18. What to Do When an Alert Appears

1. Read the alert and identify the affected food or storage location.
2. Check whether the door is closed fully and unobstructed.
3. Check current temperature and humidity readings.
4. Check for unusual gas or air-quality changes, remembering that the MQ-135 is not a spoilage test.
5. Review manufacture, expiry or best-before information, and storage duration.
6. Inspect the food using appropriate human food-safety checks.
7. Use soon only when consistent with product labels and food-safety guidance.
8. If human inspection finds the food unsuitable, follow the approved disposal procedure and update or remove the record.
9. Record notes if the implemented setup provides a notes function.

Practical responses

- **Use Soon** — verify dates and storage, prioritize the item, and inspect before use.
- **Check Food** — pause normal use, inspect the food and storage conditions, and follow food-safety guidance.
- **Door alert** — close the door, remove obstructions, wait for an update, and escalate if it repeats.
- **Unavailable sensor data** — do not treat missing data as safe; check the device and use human review while the issue is investigated.

19. Offline Operation and Internet Reconnection

When the network is unavailable, local monitoring and local alerts can continue subject to power and device operation. Remote notifications should not be expected while offline. Important unsent events may be buffered and synchronized after reconnection without creating duplicates, according to the implemented SRS behavior.

1. Verify that the device still has approved power.
2. Continue checking the LCD, LEDs, and buzzer where available.
3. Avoid repeatedly creating the same food record while the dashboard is unavailable.
4. Restore the configured network when possible without guessing credentials.
5. Wait for synchronization to complete according to the implemented setup.
6. Check the dashboard for the latest readings, records, and alerts.

Limitation — Offline operation depends on power and device operation. Continue practical food and storage checks even when no dashboard data is available.

20. Sensor Fault / Data Unavailable State

A sensor fault or unavailable reading may appear through the LCD, LEDs, buzzer, dashboard, or a status field, depending on the implemented setup. Missing data must not be treated as a safe reading.

1. Check visible connections and sensor placement only when it is safe to do so; never expose yourself to electrical hazards.
2. Check for spills, condensation, blocked airflow, or accidental movement.
3. Restart only by using the approved procedure for the supplied setup.
4. Compare local information with the dashboard when the network is available.
5. Use human review for affected freshness decisions while the reading is unavailable.
6. Contact the approved support route if the fault remains.

Safety — Do not open, repair, rewire, or handle power connections unless the approved procedure explicitly permits safe user action.

21. Daily/Weekly User Routine and Good Storage Practices

Daily checks

- ☐ Confirm that the door closes fully.
- ☐ Check that the LCD and LEDs show normal operation or a understood status.
- ☐ Check that the dashboard is updated when connected.
- ☐ Review urgent alerts and act on affected records.
- ☐ Confirm that food is visibly and safely stored.

Weekly checks

- ☐ Review Use Soon records and verify available dates and unique IDs.
- ☐ Update or remove consumed and discarded items.
- ☐ Inspect sensor placement and cleanliness without damaging sensors.
- ☐ Check visible cables and approved power connections.
- ☐ Review recurring alerts and unresolved statuses.

Good storage practices

- Follow product labels and applicable food-safety practices.
- Use suitable containers and separate raw from ready-to-eat foods.
- Avoid overpacking and maintain airflow.
- Minimize door-open time.
- Use a simple first-in/first-out habit where appropriate.

22. Troubleshooting Table

Problem / Sign	Likely user-level checks	What to do	When to escalate
No display	Approved power, visible connections, safe placement.	Use the approved power-on or restart procedure; check local indicators.	Escalate if the display remains unavailable or power handling is unsafe.
Abnormal readings	Door closure, sensor placement, airflow, condensation, containers.	Correct safe placement, wait for an update, and review the affected record.	Escalate if readings remain abnormal or conflict with the setup guidance.
Repeated alerts	Alert details, door state, dates, storage duration, recurring conditions.	Follow the alert procedure and record notes when available.	Escalate unresolved or recurring alerts after safe user checks.
Door alert	Door closure, obstructions, cables, sensor alignment or placement.	Close the door, remove obstructions, and wait for the status update.	Escalate if the door is closed but the alert continues.
No dashboard update	Network state, device power, latest data time.	Continue local monitoring and restore the configured network when possible.	Escalate if synchronization does not resume.
Network loss	Network availability and device power.	Avoid duplicate records; use local indicators and reconnect when possible.	Escalate if the approved network route remains unavailable.
Sensor fault	Visible placement, spills, condensation, safe connections.	Do not treat missing data as safe; restart only by the approved procedure.	Escalate if data remains unavailable or food decisions are affected.

23. User Safety and Important Limitations

Important limitation — FreshGuard is a prototype and monitoring aid, not food-safety certification. Do not consume food solely because the system shows Fresh/Normal.

- Follow product labels, applicable food-safety practices, and approved disposal procedures.
- Remember that thresholds vary by food category and implemented configuration.
- Understand that calibration, placement, handling, power, condensation, airflow, and network conditions affect results.
- Treat MQ-135 relative changes as an indication only; they do not confirm spoilage.
- Never interpret missing or delayed data as evidence that food is safe.
- Use human inspection and approved disposal procedures when food is unsuitable.
- Do not handle wiring or power connections unsafely.
- Keep electronics protected from liquids and unsuitable environments.

24. Quick Reference Status & Alert Table

Indicator / status	What it generally means	Immediate user action	Important reminder
Fresh/Normal	Configured conditions and timing appear within the expected range.	Continue normal checks.	Still inspect food and follow labels.
Use Soon	Attention is needed soon.	Verify dates and storage; prioritize and inspect.	Do not ignore product guidance.
Check Food	Human checking is required.	Inspect food and storage before deciding.	Not a guaranteed spoilage decision.
Green	Normal or no immediate attention indicated.	Continue routine monitoring.	Check context on the LCD or dashboard.
Yellow	Caution or attention soon.	Review the related status or alert.	Exact logic may be configured.
Red	Urgent attention or an alert.	Inspect promptly and follow the alert procedure.	Color alone does not identify the cause.
Door-open alert	Door may be open or improperly closed.	Close it and remove obstructions.	Escalate if it repeats.
Sensor fault / data unavailable	A reading cannot be relied on.	Check safely and use human review.	Missing data is not a safe reading.
Network / offline state	Dashboard or remote communication is unavailable or delayed.	Use local monitoring and reconnect when possible.	Remote notifications should not be expected offline.

25. End-User Checklist

Print this page or use it during setup and routine checks.

- ☐ Setup location chosen: refrigerator, storage area, or insulated box/cooler simulation.
- ☐ Sensors placed safely and according to the implemented setup.

- ☐ Approved power source checked.
- ☐ LCD and LED operation confirmed.
- ☐ Dashboard connection checked when required.
- ☐ Food records created with all available fields.
- ☐ QR option confirmed, or normal search by ID/name understood.
- ☐ Door closes fully without obstruction.
- ☐ Daily alert review completed.
- ☐ Weekly inventory, date, and unique-ID review completed.
- ☐ Abnormal readings handled using the alert and safety guidance.
- ☐ Offline and reconnection behavior understood.
- ☐ Safety limitations read and understood.
- ☐ Approved support contact or route recorded.

26. FAQ

Does FreshGuard decide whether food is safe? No. It supports monitoring and attention decisions; human inspection, product labels, and food-safety practices remain necessary.

What does Use Soon mean? The record or monitored conditions indicate that attention is needed soon. Verify dates and storage, prioritize the item, and inspect it before use.

What does Check Food mean? A condition, timing issue, data pattern, or unavailable reading requires human checking before a decision is made.

Does MQ-135 prove spoilage? No. It indicates relative air-quality or gas-pattern changes and does not confirm spoilage or identify a specific contaminant.

What if the internet is down? Use local monitoring and alerts if power and the device remain available. Reconnect when possible and check synchronization afterward.

What if the door alert repeats? Check closure, containers, cables, alignment, and placement. Escalate if it continues after safe checks.

What if a sensor reading is missing? Do not treat missing data as safe. Check placement and visible conditions safely, then follow the support route if the issue remains.

Can I use QR codes? Yes, where QR functionality is implemented. Otherwise search by unique ID or name and update the record normally.

Why do thresholds differ by food? Expected conditions and timing depend on the food category and the implemented configuration.

Can I use FreshGuard in a refrigerator or insulated box/cooler simulation? Yes, when the setup is arranged and used according to the implemented instructions.

What should I do with unsuitable food? Follow the approved disposal procedure and update or remove its record.

How do I update consumed or discarded items? Open or search for the record, change its state according to the implemented interface, save it, and confirm the inventory is current.

27. Support/Notes Page

Approved support contact or project link

[Insert approved support contact or project repository link]

Implemented setup notes

[Insert implemented setup notes, if approved]

Notes

When requesting help, provide:

- ☐ Device state and whether approved power is available.
- ☐ LCD or LED indication.
- ☐ Affected sensor or storage location.
- ☐ Dashboard timestamp or latest-data time.
- ☐ Recent alert message or status.
- ☐ Power and network condition.
- ☐ Steps already tried.